

MedSim VRAI — Field Test Plan & Coverage Matrix

Comprehensive field validation across all core system areas · prepared for the expanded field test

Date: 2026-06-13 · **Scope:** 72 test cases across 13 core areas (33 are P1 / must-pass) · **Companion workbook:** MedSimVRAI_Field-Test-Matrix.xlsx (execution + issue log + live coverage rollout) · **Goal:** no gaps — every core capability gets exercised on real hardware before broader rollout.

How to use this

- 1. **Pre-flight first.** Run `scripts/preflight.sh` (must be 6/6) and complete **Area A (network & pairing)** — it gates everything else.
- 2. **Work area by area** in the Excel *Test Matrix* sheet. For each row set **Result** (Pass / Fail / Blocked / Not run / N-A), your initials, the date, and notes. Don't leave any **P1** row "Not run" before sign-off.
- 3. **Coverage is visible** on the workbook's *Coverage* sheet — a live Pass/Fail rollout per area, so gaps show at a glance.

→ **Turning issues into fixes (the Claude pipeline).** When a test fails, add a row to the workbook's **Issue Log** — its columns are ordered to give Claude everything it needs:

- 1. Fill **Title** · **Steps to reproduce** · **Expected** · **Actual** · **Environment** · **Severity**, link the **Test ID**, set **Status = New**.
- 2. Filter **Status = New**, copy those rows (including the header), and paste into Claude Code prefixed with: "Fix these field-test issues:"
- 3. Claude triages, fixes, runs the gates, and replies with the commit. Retest, then set **Status = Fixed** → **Verified**.

This makes the field test a closed loop: observe → log → paste → fixed → re-verify.

Coverage map

Area	Tests	P1	Status
A · Network & pairing (FR-004)	6	4	
B · Avatar tablet (FR-006)	5	2	
C · Audio station (FR-006b)	6	3	
D · Voice loop end-to-end	4	2	
E · Control room (FR-005)	5	2	
F · Say-as-character (FR-003)	2	2	
G · Med ordering (FR-001/002)	6	3	
H · Staged errors (FR-008)	14	7	
I · EHR / Medical Records	4	2	
J · Multi-patient / room	5	2	
K · Devices	5	0	
L · Debrief	5	2	
M · Non-functional	5	2	

TOTAL	72	33	
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Already field-validated at generation: **H11** — the staged-error debrief correctly reported a MISSED error (the arm→missed→debrief arc).
Everything else awaits this matrix.

Test cases by area

A · Network & pairing (FR-004)

ID	Pri	Test case	Steps	Expected result	Device(s)	Result / notes
A1	P1	Preflight passes 6/6 Mac on session network	Run scripts/preflight.sh	All 6 checks green; QR minted; onboarding URL printed	Mac	
A2	P1	Cert SAN covers current IP Network identity known	Preflight check 3 / cert-doctor.sh	Leaf cert SAN includes the Mac's current LAN IP	Mac	
A3	P1	Root CA trusted on each tablet CA installed once per device	Open https portal on tablet	Green lock, NO cert warning, getUserMedia allowed	iPad,Android	
A4	P2	Onboarding page loads on a NEW tablet Fresh tablet, CA not trusted	Scan QR → proceed → /onboard → install CA	Page renders; CA downloads + installs; lock turns green	iPad,Android	
A5	P1	Travel-router (Beryl) full loop Tablets on Beryl, MAC-privacy OFF	Run a full voice take over the kit LAN	Loop completes; NO new trust step (CA holds across networks)	iPad,Android	
A6	P2	Mac IP stable across restart Static DHCP lease bound	Restart router/Mac; recheck IP	Mac keeps the same IP; QR/cert still valid	Mac	

B · Avatar tablet (FR-006)

ID	Pri	Test case	Steps	Expected result	Device(s)	Result / notes
B1	P1	Patient QR → 3D face loads Patient persona, avatar mode	Scan patient QR	Translucent 3D face renders; no console errors	iPad	
B2	P1	Lip-sync tracks speech Avatar live	Trigger a spoken reply	Visemes track the audio; no desync	iPad	
B3	P2	Emotion / altered-state reflected Persona w/ emotion state	Drive a turn with emotion	Expression reflects the persona's state	iPad	
B4	P3	Translucency slider works Avatar live	Drag the slider	Opacity changes smoothly	iPad	

B5	P2	Reconnect after network blip Avatar live	Drop + restore Wi-Fi	WS reconnects; avatar resumes without reload	iPad	
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C · Audio station (FR-006b)

ID	Pri	Test case	Steps	Expected result	Device(s)	Result / notes
C1	P1	Audio QR → flat portrait (no 3D) Non-patient, audio mode	Scan audio-mode QR	Flat portrait + name + badge; NO 3D rig	Android	
C2	P1	Push-to-talk records Audio station up	Hold, speak, release	Recording captured; transcription begins	Android	
C3	P1	Room-local STT (Mac) transcribes Portal up, STT warm	One spoken take	Transcript in ~1–2 s; character replies in voice	Android,Mac	
C4	P2	Drug names transcribe correctly Med board active	Say an order w/ a drug name	Drug name faithful (e.g. 'ceftriaxone')	Android,Mac	
C5	P3	Name shows (not internal ID) Bound character	Read the header	Persona name (e.g. 'Pharmacist Lee')	Android	
C6	P2	On-device backup if portal drops Portal unreachable	Take after drop	Names the failure + arms on-device fallback	Android	

D · Voice loop end-to-end

ID	Pri	Test case	Steps	Expected result	Device(s)	Result / notes
D1	P1	Full loop PTT→STT→AI→TTS Character bound, running	One complete take	Speech → transcript → character voice reply	iPad,Android	
D2	P1	First words within target Warm session	Time a reply	First audio ~2–3 s (streaming)	iPad,Android	
D3	P2	iPad on-device STT (WebGPU) iPad avatar	Take on iPad	On-device whisper (~1–1.5 s warm)	iPad	
D4	P2	Several takes in a row Warm session	5 takes back-to-back	Consistent latency; no degradation	iPad,Android	

E · Control room (FR-005)

ID	Pri	Test case	Steps	Expected result	Device(s)	Result / notes
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E1	P1	Setup page renders all cards Logged in, configured	Open /portal/control/setup	Med, devices, skins/voices, add-character present	Mac	
E2	P1	START opens Live Ops window On Setup	Press ► START	Goes live; Live Operations opens in a NEW window	Mac	
E3	P2	Add character mid-session Live session	Add a persona via banner	Character appears + becomes scannable	Mac	
E4	P2	Avatar ⇌ audio toggle Setup/Live	Flip a character's mode	QR re-mints for the chosen mode	Mac	
E5	P3	Configured session lands on Setup Configured	Navigate to /ops directly	Redirects to Setup	Mac	

F • Say-as-character (FR-003)

ID	Pri	Test case	Steps	Expected result	Device(s)	Result / notes
F1	P1	Verbatim mode Character bound	Say-as: verbatim line	Character speaks the exact text	iPad,Android	
F2	P1	In-character mode Character bound	Say-as: in-character intent	AI rephrases in persona voice/emotion	iPad,Android	

G • Med ordering (FR-001/002)

ID	Pri	Test case	Steps	Expected result	Device(s)	Result / notes
G1	P1	Doctor orders decisively WITH dose Doctor, condition set	Ask the doctor for orders	Doctor PLACES an order w/ a full dose; no 'what's available'	iPad,Android	
G2	P2	Doctor escalates if already-on Patient already on primary	Ask again	Doctor picks a different primary / escalates	iPad,Android	
G3	P1	Instructor flags a med unavailable Med board open	Toggle a med unavailable	Med marked unavailable in board state	Mac	
G4	P1	Pharmacist coaches an alternative Med flagged unavailable	Ask the pharmacist	Names an AUTHORED alternative (never invented)	Android	
G5	P2	Doctor→pharmacist→doctor loop Above set up	Run the loop	Doctor approves/adjusts w/ a dose	iPad,Android	
G6	P2	Board condition matches the patient Single + multi	Open the med board	Condition/formulary matches the PATIENT	Mac	

H • Staged errors (FR-008)

ID	Pri	Test case	Steps	Expected result	Device(s)	Result / notes
H1	P1	Builder wizard flows end-to-end Session running	Walk all 6 steps; pick a candidate	Next enables on selection; review; ARM works	Mac	
H2	P1	Transcription (verbal) deliver→correct Sound-alike armed (verbal)	Question, then read-back challenge	Doctor speaks wrong drug, defends, then CORRECTS + credits the catch	iPad,Android	
H3	P1	Wrong dose (document) discrepancy Armed wrong_dose-document-charting	Trainee reviews chart	Note contradicts the MAR; MAR stays correct (catchable)	iPad	
H4	P2	Interaction error Armed interaction	Run the encounter	Conflicting med vs an active med; pharmacist coaches	iPad,Android	
H5	P2	Allergy oversight Documented allergy present	Armed allergy	Allergy-conflicting med ordered despite the allergy	iPad,Android	
H6	P2	Administration error Armed admin-document	Trainee preps meds	MAR shows expired/wrong-time/look-alike deviation	iPad	
H7	P1	Patient impact trigger + stabilize Armed w/ impact	Trigger; then Stabilize	Patient symptomatic + vitals shift; Stabilize reverts both	iPad,Mac	
H8	P2	Auto-trigger on administration Impact = on_administer	Administer the staged med	Impact fires automatically	Android,Mac	
H9	P2	Severe impact double-confirm Severe impact armed	Press Trigger	Demands a second confirmation before firing	Mac	
H10	P1	Caught / Missed marking Error armed	Mark caught, then missed	Outcome recorded; transcript stamped	Mac	
H11	P1	Debrief shows the full arc Resolved error(s)	Open the debrief	Arc armed→delivered→impact→resolved + outcome (MISSED highlighted) [partial 	Mac	
H12	P2	Disarm restores chart byte-exact Document error armed	Disarm it	Mutated chart artifact restored exactly	Mac	
H13	P1	Multi-patient per-bed isolation 2–3 bed room	Arm on bed A only	Bed A shows it; bed B unaffected; build scoped to bed	Mac	
H14	P3	Builder opens in its own window Setup/console	Click Build a staged error	Separate window; closing returns to the control panel	Mac	

ID	Pri	Test case	Steps	Expected result	Device(s)	Result / notes
I1	P1	MR shows the correct PATIENT	Open the Medical Record	Chart is the PATIENT (never the doctor)	Mac	

		Single + multi				
I2	P2	MR opens in its own window Control panel	Launch EHR	Separate window; closing returns to an intact panel	Mac	
I3	P1	Chart content correct MR open	Inspect MAR/allergies/vitals/notes	All present + patient-correct	Mac	
I4	P2	Order entry works MR open	Place a lab/imaging/med order	Recorded + visible in chart/debrief	Mac	

J • Multi-patient / room

ID	Pri	Test case	Steps	Expected result	Device(s)	Result / notes
J1	P1	Start a 2-3 bed room Room wizard	Finalize a room	Each bed gets a join code + console	Mac	
J2	P1	Per-bed roster + patient correct Room running	Open each bed console	Correct patient + FULL roster checked (doctor/charge/RT/family)	Mac	
J3	P2	Per-bed tools scoped Room running	Use board/errors/devices on a bed	Each tool scopes to that bed only	Mac	
J4	P3	Nursing station supervises beds Room running	Open the nursing station	All beds' strips/transcripts visible	Mac	
J5	P2	Transcripts land on the right bed Multi-bed turns	Take on two beds	Each transcript lands on its own encounter	Mac,iPad,Android	

K • Devices

ID	Pri	Test case	Steps	Expected result	Device(s)	Result / notes
K1	P2	Device join via QR Device skin set	Scan a device QR	Device station joins + renders	iPad,Android	
K2	P3	Pump program / alarm Pump joined	Program a rate / alarm	Pump reflects program; alarm propagates	iPad	
K3	P2	Med cart administer logged Cabinet joined	Administer a med	'RN administered ...' logged to transcript	iPad,Mac	
K4	P2	Telemetry + instructor override Monitor up	Override a metric	Bedside monitor reflects the override	iPad,Mac	
K5	P2	Scene inject Live encounter	Inject vitals.drop / code.blue	Scene fires; vitals/alarms update	Mac	

L • Debrief

ID	Pri	Test case	Steps	Expected result	Device(s)	Result / notes
L1	P1	Debrief builds after session Session w/ activity	End session; open debrief	Renders w/ summary + transcript	Mac	
L2	P2	Paired turns + tagging Debrief open	Inspect transcript	Turns paired; NCJMM tagged	Mac	
L3	P1	Staged-error arc section Errors resolved	Inspect debrief	Full staged-error arc + outcomes	Mac	
L4	P3	Documentation / orders alignment Charting locked	Inspect debrief	Alignment cards populate	Mac	
L5	P2	Per-patient debrief (multi) Room session	Open each bed's debrief	Each patient's debrief separate + correct	Mac	

M • Non-functional

ID	Pri	Test case	Steps	Expected result	Device(s)	Result / notes
M1	P1	Audio never leaves device/room Any voice take	Confirm STT routing	On-device (iPad) or room Mac only; no third party	iPad,Android,Mac	
M2	P2	Sustained 20-min session iPad avatar	Run ~20 min continuous	No thermal throttle / latency creep	iPad	
M3	P2	Reconnection after blip Live session	Drop + restore network	Stations reconnect; session continues	iPad,Android	
M4	P3	Device token enforcement (if on) MEDSIM_FACE_TOKEN=1	Hit /listen or /stt w/o token	403 rejected	Mac	
M5	P1	Restart behavior understood Mid-session	Restart the portal	Known: session wiped → fresh Setup (FR-010 will improve)	Mac	